Unit 5 Test - ANSWERS

MODIFIED TRUE/FALSE z

1. ANS: F, an increase

2. ANS: F, bottom right side

3. ANS: T

4. ANS: F, negative

5. ANS: F, hydrogen gas at the cathode and oxygen gas at the anode

MULTIPLE CHOICE

6. ANS: D

7. ANS: B

8. ANS: C

9. ANS: A

10. ANS: B

11. ANS: A

12. ANS: D

13. ANS: A

14. ANS: B

15. ANS: A

What is oxidized and what is reduced in the following reaction: $N_2 + 2O_2 \rightarrow 2NO_2$
N_2 is oxidized; O_2 is reduced

Determine the oxidation number of each element in the following substances:
a) H_2SO_4 H:+1, S:+6; O:-2 b) $Ca_3(PO_4)_2$ Ca: +2, P: +5; O:-2 c) Cl_2 Cl: 0
d) $KMnO_4$ K:+1, Mn:+7, O:-2 e) H_3AsO_4 H: +1; As: +5, O:-2 f) $CaSnO_3$ Ca:+2, Sn: +4, O:-2

Balance the following equations (use the oxidation number method):

a. $MnO_4^- + 5\,Fe^{2+} + 8\,H^+ \rightarrow Mn^{2+} + 5\,Fe^{3+} + 4\,H_2O$ (acidic solution)
b. $2\,HNO_3 + 3\,H_2S \rightarrow 2\,NO + 3\,S + 4\,H_2O$
c. $Cr_2O_7^{2-} + 6\,Cl^- + 14\,H^+ \rightarrow 2\,Cr^{3+} + 3\,Cl_2 + 7\,H_2O$ (acidic solution)
d. $2\,Cr(OH)_3 + ClO_3^- + 4\,OH^- \rightarrow 2\,CrO_4^{2-} + Cl^- + 5\,H_2O$ (basic solution)
e. $Zn\,(s) + SO_4^{2-} + 4\,H^+ \rightarrow SO_2\,(g) + Zn^{2+} + 2\,H_2O$

Iodine normally has oxidation numbers of -1 and 0. Could I_2 serve as a reducing agent? Explain.
This is not likely to happen because for I_2 to be oxidized it would have to lose an electron forming a +1 oxidation number which iodine does not typically do.

Describe what happens when chlorine in water is mixed with bromide ions and mineral oil (a non-polar solvent). Explain why this happens.
The mineral oil layer will turn a golden yellow layer indicating that Br_2 is produced.
Net reaction: $Cl_2 + 2\,Br^- \rightarrow 2\,Cl^- + Br_2$

Should chlorine gas be passed through copper tubing? Explain your answer using E° values.
No, E°_{cell} = 1.02 V which indicates a spontaneous reaction. If the reaction is spontaneous then the copper tubing would be oxidized and eventually disappear.

Balance the following equations using the half-reaction method.

f. $Ba\,(s) + Cl_2 \rightarrow Ba^{2+} + 2\,Cl^-$
g. $MnO_4^- + 5\,NO + 2\,OH^- \rightarrow Mn^{2+} + 5\,NO_2^- + H_2O$
h. $2\,I^- + 2\,NO_3^- + 4\,H^+ \rightarrow I_2 + 2\,NO_2 + 2\,H_2O$

For each set of reactants, write an equation for the reaction. Use E° values to determine if the reaction is spontaneous.

i. $Mg_{(s)} + Al^{3+}$

$3\ Mg_{(s)} + 2\ Al^{3+}_{(aq)} \rightarrow 3\ Mg^{2+}_{(aq)} + 2\ Al_{(s)}$ $E^o_{cell} = +0.71\ V$ Spontaneous

j. $Hg^{2+} + Ag_{(s)}$

$Hg^{2+}_{(aq)} + 2\ Ag_{(s)} \rightarrow Hg_{(l)} + 2\ Ag^{+}_{(aq)}$ $E^o_{cell} = 0.05\ V$ Spontaneous

k. $Cl^{-} + I_{2\,(s)}$

$2\ Cl^{-}_{(aq)} + I_{2\,(s)} \rightarrow 2\ Cl_{2\,(g)} + 2\ I^{-}_{(aq)}$ $E^o_{cell} = -0.82\ V$ Not spontaneous

l. $Sn_{(s)} + H^{+}$

$Sn_{(s)} + 2\ H^{+}_{(aq)} \rightarrow H_{2\,(g)} + Sn^{2+}_{(aq)}$ $E^o_{cell} = +0.14\ V$ Spontaneous

m. $H_{2(g)} + Cl_{2(g)}$

$H_{2\,(g)} + Cl_{2\,(g)} \rightarrow 2\ H^{+}_{(aq)} + 2\ Cl^{-}_{(aq)}$ $E^o_{cell} = +1.36\ V$ Spontaneous

n. $Cu_{(s)} + I^{-}$

Both $Cu_{(s)}$ and $I^{-}_{(aq)}$ can only be oxidized. Therefore no redox reaction can occur.

Name two substances which oxidize Ag to Ag^{+} .

Any substance above and on the left side of the redox table from Ag(s) (e.g. Hg^{2+}, Br_2)

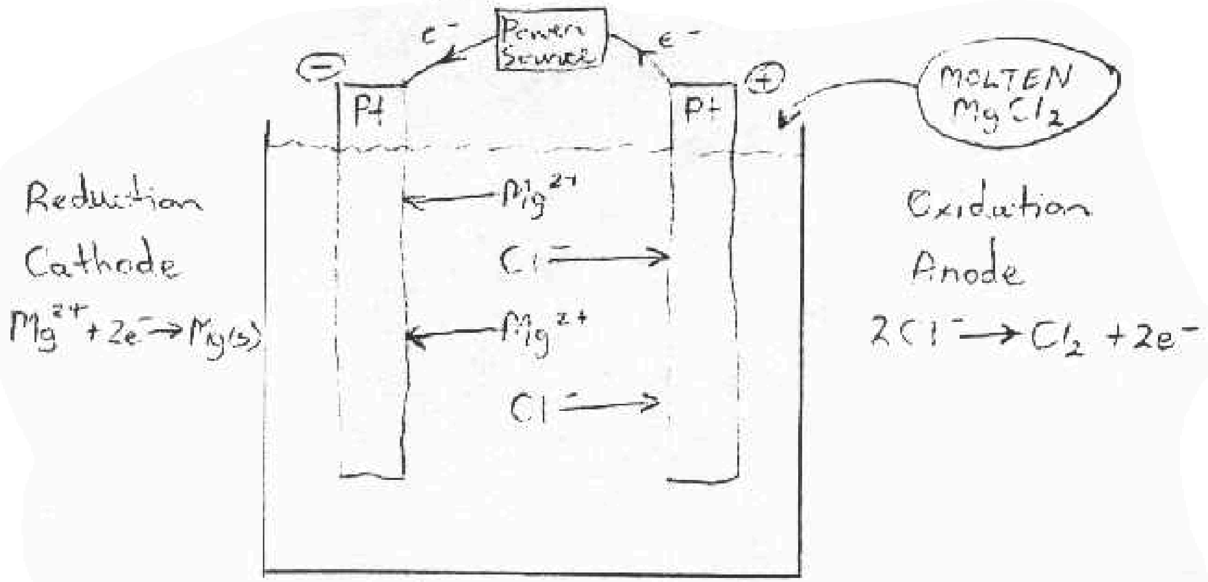
A current of 156A is applied to an electrolytic cell containing molten $CuCl_2$ for 45 min. What substance and how many grams of it will form on the cathode?

1.4×10^{2} g of copper will form at the cathode.

How many moles of electrons must have been added to an electrolytic cell containing Cd^{2+} ions if 10.5g of Cd(s) forms on the cathode?

0.187 mol e^{-}

Draw a diagram of an electrolytic cell for the electrolysis of molten $MgCl_2$. Label all parts of the cell and indicate the reactions occurring at each electrode. Estimate the voltage required to electrolyze $MgCl_2$.



$E^\circ = -2.37 - 1.36 = -3.73\,V$ ∴ at least 3.73 V would have to be applied.